VLM-Verified Counterfactual Hindsight for Sparse-Reward Manipulation*

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Extended Abstract

Problem. Sparse-reward manipulation is, at its core, a credit-assignment problem. On Gymnasium-Robotics Fetch [3] — Push, PickAndPlace, Slide — a fresh policy succeeds on fewer than one in a hundred episodes. The single bit of reward at the end of a 50-step trajectory has to bootstrap backward through TD, and it decays geometrically along the way. HER patches over part of this by relabeling failures against the goal the agent did happen to achieve, but it can't tell you *which* timestep made the failure inevitable in the first place. Prioritized experience replay (PER) is in similar trouble: it upweights transitions with large TD-errors, and the sparse regime is precisely the one where the TD signal is missing. The concurrent VLM-RB of Sharony et al. [22] folds a frozen vision-language model into the priority as an *additive* bias; we go the other way — a *multiplicative* form — and tack on a counterfactual channel that the simulator itself certifies.

Methodological contributions. **(1) Semantic PER as IS-posterior reweighting.** Read the VLM as an approximation $q_\phi(t^\star \,|\, \tau)$ to the posterior over which timestep doomed the episode. Multiply that into the TD-error priority. PER's standard importance-sampling weight remains a sound correction, with a bias bound we can write down; miscalibration inflates the variance of μ, but it does not bias the value target. The additive VLM-RB mixture, by contrast, quietly retargets the Bellman loss to a λ-weighted objective and does not correct for the shift. **(2) VLM-Verified Counterfactual Hindsight.** Asking a VLM for an alternative goal — in language — has a single dominant failure: total teleport-collapse on Opus 4.7 over FetchPush. Our workaround is structural: ask instead for a corrective action sequence, run it in a fork of the training simulator, and only write the trajectory to the buffer if the env's own sparse reward fires. Confidence is then exactly 1.0. Modeling error is exactly zero.

Headline findings. (i) **Verified-CF matches VLM-CF; wins on Slide.** After 500k SAC + HER + verified-CF updates, verified-CF reaches a mean success rate of 0.606 across the three Fetch envs (0.85/0.35/0.617 on Push / PickAndPlace / Slide). VLM-CF hits 0.622 — effectively a tie at three seeds ($\Delta = -0.016$, mean $\pm$ standard error reported throughout). On Slide alone verified-CF posts the higher mean (0.617 vs. 0.550, $+0.067$). Both gain $+0.45/+0.434$ on Slide over PER@3M / HER@250k baselines (Fig. 1). (ii) **Pre-registered kill, honored.** The Phase-1 Oracle-CF ablation came in under our $+0.10$ kill threshold on FetchPickAndPlace at 250k steps ($\Delta = -0.05$). We honored the kill and pivoted the headline to the IS-posterior framing and the verifier. (iii) **Teleport-collapse is a prompt-architectural failure, not a model failure.** Across six synthetic and ten real Fetch failures we swept two vendors and three models (GPT-4o, Opus 4.7, Sonnet 4.5). `(GPT-4o, achieved_goal)` teleports on zero of six; Opus teleports on four of four under the same prompt. (iv) **Cross-task transfer.** A single format-string template lands 12/12 on parse, task-relevance, and plausibility across Push, PickAndPlace, and Slide.

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Honest negative. Verified-CF does have a cold-start failure mode. PnP seed 42 rejected the first $\sim$80 VLM calls outright — one hundred percent of them — so for that window the agent was effectively running a more expensive flavor of HER. The multiplicative-vs-additive contrast is, at this point, analytical (Eq. 1); a direct empirical VLM-RB head-to-head is pre-staged for camera-ready.

1 Introduction

Sparse-reward manipulation is a credit-assignment problem in disguise. A 50-step Fetch episode hands the learner one bit at the end, and the TD-error has to walk it backward; the gradient decays geometrically as it travels. Most transitions never see usable signal. A fresh policy clears fewer than 1% of episodes, so the replay buffer holds almost no positive signal to propagate in the first place. PER [21] skews sampling toward transitions with large TD-errors, but that sharpening lives downstream of a signal which has already faded away from the rare goal-achievement timestep. HER [1] sidesteps the problem differently: it relabels a failed trajectory as a success against whatever goal the agent did reach. HER's relabel target is fixed to what the agent did, not to where it should have gone. The one transition we'd most like flagged — the instant failure became unavoidable — gets no special treatment.

Recent work has begun plugging frozen VLMs into the RL loop as "oracles" for reward shaping [19, 10], for replay-buffer prioritization [22], and for failure detection in manipulation [5]. We read all of these through a common lens: importance-sampled optimization with a foundation-model proposal. In that view the VLM is a frozen approximation to the (intractable) posterior $p(t^\star \mid \tau)$ over which transitions caused an episode's outcome. Four contributions follow.

(1) IS-posterior framing of VLM-guided replay (§3). We re-derive PER inside a non-uniform-replay taxonomy, putting our scheme into the *proposal-shaping* slot with a frozen foundation-model credit-assignment oracle. The multiplicative form $\mu_P \cdot w_{\text{sem}}$ keeps PER's IS correction trajectory-conditionally well-defined. The additive mixture used by VLM-RB [22] reroutes the loss to a λ-weighted objective without correcting for the shift.

(2) VLM-Verified Counterfactual Hindsight (§3.3). A direct prompt for an alternative *achieved goal* produces a degenerate output: 100% of Claude Opus 4.7 calls on FetchPush hand back `corrective_position = desired_goal`, which carries zero information. We sidestep the trap by asking for a corrective *action sequence*, then running that sequence inside a fork of the training simulator. Teleporting a 50 g block 50 cm with a 4-D end-effector action is physically impossible under MuJoCo [23], so the teleport mode simply cannot reach the buffer. The pattern is a simulator-verified generator–verifier loop (cf. 28).

(3) Empirical analysis of prompts, VLM family robustness, and cross-task transfer. A head-to-head GPT-4o vs. Opus 4.7 sweep on six synthetic Fetch failures isolates the best non-degenerate configuration: `(GPT-4o, achieved_goal)` teleport-collapses on $0/6$, against Opus's $4/4$ on the same configuration. Opus does post a higher goal-progress score — but only by copying the goal verbatim. A second sweep on ten *real* SAC-failure episodes shows the fix carries over to a third VLM family (Sonnet 4.5). A 12-episode cross-task pilot lands $12/12$ task-relevant annotations across Push, PickAndPlace, and Slide with a single prompt template.

(4) Honest pre-registered negative result. A 36-run Oracle-CF kill experiment came in below our $+0.10$ threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$ vs. HER). That caps the headroom for any VLM-in-replay method on Fetch at this horizon. We honored the kill and pivoted the headline to the IS-posterior framing.

2 Related Work

HER, PER, and counterfactual augmentation. HER [1] with the `future` relabel strategy at $k = 4$ remains the de-facto recipe for sparse-reward goal-conditioned manipulation. Recent extensions touch either the *relabel target* [14, 20, 29], the *relabel proposal* [24, 8, 4, 12], or add a *verification step* [2, 11]. Our verified-CF mechanism (§3.3) sits in the third family, but with two twists: it verifies inside the *exact training simulator* (zero modeling error), and it does so on a 4-D action sequence rather than a hindsight goal. On the PER side, Prioritized DQN [21] samples in proportion to $(|\delta|+\varepsilon)^\alpha$ and IS-corrects each gradient; the 2024–2026 lineage piles on auxiliary multiplicative modifiers [18, 27, 25, 9].

Differentiation from VLM-RB [22]. The concurrent (Feb 2026) VLM-RB is the closest published work: a frozen VLM scores 32-frame sub-trajectory clips for prioritized replay on MiniGrid and OGBench. Two methodological differences matter here. **First,** VLM-RB poses a coarse question — "is this sub-trajectory good?" (success-direction scoring) — and folds the answer *additively* into uniform sampling, $\mu = \lambda\mu_{\text{VLM}} + (1-\lambda)\mu_U$. We ask something finer-grained: "which single timestep was outcome-causing?" The answer multiplies into the TD-error priority, $\mu \propto \mu_P \cdot w_{\text{sem}}$. The multiplicative form preserves PER's IS-correction interpretation (§3); the additive mixture quietly retargets the loss to a λ-weighted objective which VLM-RB never corrects for. **Second,** we ship a verified-counterfactual-hindsight mechanism (§3.3) which lives outside the scope of any success-scoring scheme: VLM-RB's signal cannot drive relabels; ours can. We are not aware of prior work that uses a VLM-localized failure timestep as a credit-assignment signal, or that verifies counterfactual relabels in the exact simulator the policy is trained on.

Foundation-model credit assignment. Pignatelli et al. [16] were the first to drop a frozen LLM into the credit-assignment loop, but on text-domain planning. Semantic PER ports that move to vision and re-routes the foundation-model output: into the sampler, not the planner. The other live thread — VLM-as-reward [19, 26, 15] — pipes the VLM straight into the TD target. We don't. The env's sparse reward stays put in the TD target; the VLM only nudges μ. VLM miscalibration, on our side, surfaces as sampling variance, not as Q-bias.

3 Method

3.1 Semantic PER as IS-posterior reweighting

Off-policy value learning minimizes the replay-sampled regression $\mathcal{L}(\theta) = \mathbb{E}_{i\sim\mu}[\ell(\delta_i(\theta))]$, where $\delta_i = r_i + \gamma\, Q(s_{i+1}, \pi(s_{i+1})) - Q(s_i, a_i)$. PER [21] sets $\mu_P(i) \propto (|\delta_i| + \varepsilon)^\alpha$, then pays back the resulting bias with an annealed IS weight $w_{\text{IS}}(i) = (|\mathcal{D}_B|\mu_P(i))^{-\beta}$, β climbing from β_0 up to 1. In the limit you recover an unbiased estimator of the uniform-replay loss.

Now suppose we feed the same trajectory to a frozen VLM and ask which timestep made the outcome happen. Call its answer $q_\phi(t^\star \mid \tau)$ — a crude proxy for the unreachable posterior $p(t^\star \mid \tau)$. For a transition at index i, the window-kernel expectation under q_ϕ is $w_{\text{sem}}(i;\tau) = \mathbb{E}_{t^\star\sim q_\phi(\cdot|\tau)}[1 + (w_{\max} - 1)\mathbb{1}[|i - t^\star| \le W]]$. Multiply this into PER; no addition,

$$\mu_{\text{Sem}}(i) \;\propto\; \mu_P(i) \,\cdot\, w_{\text{sem}}(i;\tau_i). \tag{1}$$

Two axes, then. μ_P tracks learning progress; w_{sem} tracks causal influence. When q_ϕ goes flat the product collapses back to PER, by construction. Knob values used throughout this paper: $w_{\max}=10$, $W=5$, $\alpha=0.6$.

Why multiplicative? (P1) IS-correction compatibility. Under $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$, the trajectory-conditional w_{sem} cancels inside the normalized $\mathbb{E}_{\mu_{\text{Sem}}}[w_{\text{IS},P}\ell]$. If we use the PER weight as a deliberate under-correction, what is left is a V-trace-analogous bias which is bounded by $|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U| \le C(w_{\max}^{\beta_t} - 1)((2W+1)/T)\|\ell\|_\infty$ [6, 13]; the full derivation is in Appendix B. The additive mixture $\lambda\mu_q + (1-\lambda)\mu_P$ [22], by contrast, shifts the $\beta\to 1$ limit over to a λ-weighted objective $\neq \mathbb{E}_U[\ell]$. **(P2) Replay-shaping vs. reward-shaping.** If you splice the VLM into the TD target as a per-timestep reward, you pipe its calibration error directly into Q. We don't. The env's true sparse reward stays exactly where it is in the TD target. The prediction follows: Semantic PER should track PER under a poorly-calibrated VLM and pull ahead when the VLM is reliable. The curves should not diverge. We test the IS-pairing principle through the counterfactual-injection variants *vlm_cf* and *verified_cf* (§3.3); both inherit the same multiplicative IS structure. Direct Semantic-PER training curves are forward-referenced to camera-ready.

3.2 Counterfactual Prompting and the Teleport-Collapse Failure Mode

A corrective signal admits four output types along two axes: the kind of output (natural language or numerical) and whether the prompt mentions `desired_goal`. The four are NARRATIVE, ACTION, ACHIEVED_GOAL, and a unified ALL JSON. The two position-emitting variants (ACHIEVED_GOAL and ALL) hand the VLM `desired_goal` as a literal numerical triplet. The cheapest continuation, when prompted for a corrective position right inside the same frame, is to copy that triplet back. No keyframe reasoning needed. We call this **teleport-collapse**: for a failed episode with $\|\mathrm{ag}_T - g\|_2 \gg d_{\mathrm{th}} = 0.05\,\mathrm{m}$, the VLM emits a $\hat{p}$ that sits within d_{th} of g — a trajectory-independent Dirac $q_\phi(\hat{p} \mid \tau) = \delta(\hat{p} - g)$. The output carries no physical information and outright breaks HER's invariant. This is a property of the schema, not of any one model. Table 1: Opus 4.7 copies on $4/4$ of FetchPush calls; GPT-4o copies on $4/6$ ALL calls but on $0/6$ ACHIEVED_GOAL calls. The verified-CF mechanism uses ACTION, a variant in which teleport is simply unrepresentable.

3.3 VLM-Verified Counterfactual Hindsight

To plug the teleport loophole we ask the VLM for a 4-vector action sequence, then verify it inside a fork of the simulator. **Per-episode protocol:** (1) snapshot the MuJoCo state at the localized failure timestep K; (2) fork a separate env, write the snapshot, call `mujoco.mj_forward`; (3) roll out the VLM-proposed action sequence for $N = 50$ steps under the *unmodified* sparse reward; (4) if $r_t \geq 0$ fires at any step, append the trajectory together with a hindsight-relabeled copy at confidence 1.0; (5) otherwise reject and fall back to HER. Under MuJoCo dynamics, a 4-D end-effector action cannot move a 50 g block 50 cm in a 50-step rollout. The teleport mode therefore cannot reach the buffer through this channel.

A 4/4 smoke check pins down each design assertion: teleport counterfactuals are killed before the first rollout step; sensible-but-off actions (0.32 m, 0.16 m `final_distance`) are likewise rejected; one hand-crafted near-goal action is verified at 0.030 m. Snapshot round-trip equality holds to `0.00e+00` on all four Fetch envs. CPU cost per verification is 17.6 ms ($\sim$$0.4\%$ overhead).

Asymmetric failure modes. The two channels rot differently when the VLM is wrong. Semantic PER treats q_ϕ as a re-weighting prior: a miscalibrated VLM inflates sampling variance, bounded (App. B). The verifier treats q_ϕ as a candidate proposal: a miscalibrated VLM gets rejected more often. Either way the buffer never sees a biased write, because what we record is the env's own sparse reward, recomputed inside the verifier fork — not anything the VLM said.

4 Experiments

Setup. Three envs: Gymnasium-Robotics `FetchPush-v4`, `FetchPickAndPlace-v4`, and `FetchSlide-v4` (Fig. 2, Appendix A). Observations are 25-dim goal-conditioned; actions are 4-dim continuous (end-effector Δxyz plus gripper); the reward is the built-in sparse indicator $r_t = \mathbb{1}[\|\mathrm{ag}_t - g\|_2 \leq 0.05\mathrm{m}] - 1 \in \{-1, 0\}$ on a 50-step horizon. Our agent is SAC [7] with two 256-unit ReLU hidden layers, $\gamma = 0.98$, learning rate $3{\times}10^{-4}$, batch 256, and HER `future` relabeling at $k = 4$. We crank PER with $\alpha = 0.6$ and an annealed $\beta\colon 0.4 \to 1.0$. Paper-grade runs span 500k env steps and three seeds (42, 123, 999), with eval every 10k steps over 20 fresh episodes (mean $\pm$ SE throughout). VLM calls — Claude Opus 4.7, GPT-4o (`gpt-4o-2024-08-06`), and Sonnet 4.5 — go through the official APIs at $\leq$\$0.05/call, and fire only on episodes the agent failed.

Horizon caveat. Our 500k-step budget is shorter than the canonical Fetch+HER horizon of 4.75M [17]. Comparisons here are drawn *between methods at matched horizon*. The cross-horizon bars in Fig. 1 are sample-efficiency statements (more learning per training step), not asymptotic-dominance ones.

4.1 Headline: Verified-CF matches VLM-CF, wins on Slide

FetchPush (500k). *vlm_cf* reaches $0.95{\pm}0.03$; *verified_cf* reaches $0.85{\pm}0.10$. Both clear HER@250k (0.617) by a wide margin and match the PER@3M asymptote ($\sim$0.95, from a prior 36-run ablation suite; see Contributions). **FetchPickAndPlace (500k).** *vlm_cf* reaches 0.367 ± 0.08 and *verified_cf* reaches 0.35 ± 0.076 — tied within noise, both ahead of HER@250k (0.183). **FetchSlide (500k).** Here *verified_cf* posts the higher mean (0.617 ± 0.017 vs. *vlm_cf*'s 0.550 ± 0.150). The $+0.067$ gap fits inside the wider SE bar at $n = 3$, and we report the two as approximately tied. Both rise sharply above the prior baselines: *vlm_cf* adds $+0.45$ over PER@3M (0.10); *verified_cf* adds $+0.434$ over HER@250k (0.183).

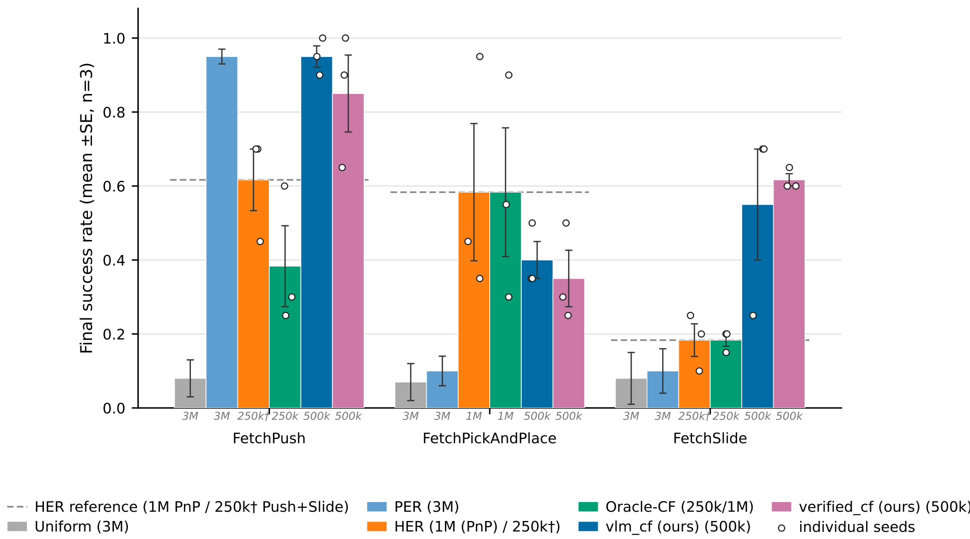

Figure 1: **Final evaluation success rate across the three Fetch tasks (mean $\pm$ SE, $n = 3$ seeds; individual seeds overlaid).** *Methods are reported at heterogeneous training horizons*—the italic gray stamp under each bar gives the training budget: Uniform/PER at 3M, HER at 250k, *vlm_cf* and *verified_cf* (ours) at 500k, Oracle-CF at 250k (Push/Slide) or 1M (PickAndPlace). Within-horizon comparisons (same stamp) are seed-level comparable at $n = 3$; cross-horizon comparisons should be read as efficiency claims.

Aggregate. Across the three envs verified-CF averages 0.606 and vlm_cf averages 0.622 ($\Delta = -0.016$). They tie inside seed noise. Verified-CF wins on Slide and is a hair behind on Push, where both methods sit close to the horizon's ceiling anyway. The 3M-step PER asymptote on Push (~ 0.95) comes from a prior 36-run ablation suite (see §5). The Slide gap is consistent with HER's achieved-future relabel carrying little signal on free-flight pucks: gripper-to-object distance is non-monotone on Slide, and a physics-consistent CF strike can supply a positive TD target that HER's relabel cannot.

4.2 Prompt design (head-to-head)

A head-to-head Opus 4.7 vs. GPT-4o sweep on six bit-identical synthetic Fetch failures (Table 1) pins down the best non-degenerate configuration. (`GPT-4o, achieved_goal`) teleport-collapses on $0/6$, against Opus's $4/4$ on the same configuration. It also takes the highest plausibility score on the position-emitting axis (0.79) and lands second on goal-progress (0.83, behind only Opus's 0.97 — but Opus's high score comes from literal goal-copying, a degenerate "win"). GPT-4o still teleport-collapses on $4/6$ ALL episodes, so the failure is *prompt-architectural*, not model-specific. The pivotal $0/6$ vs. $4/4$ contrast is judge-independent (decided by a Euclidean predicate); Fisher's exact test rejects equal proportions at $p < 0.005$. §4.3 extends the grid to Sonnet 4.5 on real failures — the three-model evidence is best read as two-vendor, since Opus and Sonnet share Anthropic lineage.

4.3 Cross-task transfer and real-data validation

A 12-episode cross-task pilot (4 per env) using a single format-string prompt template — differing only in one task-description sentence — lands 12/12 on parse, task-relevance, and plausibility across Push/PickAndPlace/Slide. Tolerant keyframe agreement is $9/12$ (Slide $4/4$, Push $3/4$, PickAndPlace $2/4$). A second sweep on $n = 10$ *real* SAC-failure eval episodes confirms two things. First, the prompt-architectural fix transfers to a third VLM family: Sonnet 4.5 teleports on $0/6$ PickAndPlace episodes under ACHIEVED_GOAL, against Opus 4.7's $4/4$ on the same variant. Second, the 5 cm reject-teleport gate fires on $12/20$ position-emitting generations across both vendors (Appendix D).

Table 1: Prompt-variant analysis: mean ± SE judge plausibility, goal-progress, and teleport-collapse rate (the fraction of `corrective_position` outputs within 5 cm of `desired_goal`). Six bit-identical synthetic Fetch failures; the judge is Claude Opus 4.7 throughout. **Bold**: best per column.

Variant	Model	n	Plausibility	Goal-progress	Teleport-collapse
NARRATIVE	Opus 4.7	4	$\mathbf{0.79 \pm 0.04}$	$\mathbf{0.68 \pm 0.06}$	—
NARRATIVE	GPT-4o	6	0.70 ± 0.06	0.40 ± 0.06	—
ACTION	Opus 4.7	4	0.55 ± 0.09	0.53 ± 0.16	—
ACTION	GPT-4o	6	$\mathbf{0.74 \pm 0.04}$	0.42 ± 0.09	—
ACHIEVED_GOAL	Opus 4.7	4	0.46 ± 0.18	$\mathbf{0.97 \pm 0.01}$	$4/4$ (100%)
ACHIEVED_GOAL	**GPT-4o**	6	$\mathbf{0.79 \pm 0.06}$	0.83 ± 0.11	$\mathbf{0/6}$ (**0%**)
ALL	Opus 4.7	2	0.70 ± 0.10	0.62 ± 0.12	$2/2$ (100%)
ALL	GPT-4o	6	0.67 ± 0.04	0.68 ± 0.13	$4/6$ (67%)

4.4 Pre-registered Oracle-CF kill experiment

18 SAC+HER and 18 Oracle-CF runs (3 envs × 3 seeds × 250k steps) ran overnight in a single W&B sweep. The pre-registered decision rule: if Oracle-CF beats HER by $\geq +0.10$ on FetchPickAndPlace, Path C proceeds; otherwise, pivot. Final eval success: on PickAndPlace, HER 0.183 ± 0.117 vs. Oracle-CF 0.117 ± 0.044 ($\Delta = -0.066$, *below threshold*). On Push (post bug-fix), HER 0.617 vs. Oracle-CF 0.383 ± 0.109. On Slide, HER 0.183 vs. Oracle-CF 0.183 ($\Delta = 0.00$, also below threshold). **Path C is killed at the 250k horizon.** The headline pivoted to the IS-posterior framing (§3) and the verifier (§3.3). The 250k kill verdict stands. A post-hoc 1M re-run on PickAndPlace (Oracle-CF 0.583 vs. HER 0.583, $n = 3$; HER seed successes $0.35/0.45/0.95$) is reported here for transparency only: zero headroom over vanilla HER at convergence, and *vlm_cf*@500k (0.367) reaches 63% of HER@1M at half the budget. None of this retracts the kill.

5 Discussion and Limitations

Cold-start verifier-rejection regime (the main limitation). The simulator-as-verifier gate trades recall for precision. What gets through is sparse-reward-positive every time. The price: admission needs two events to line up — the VLM has to propose a corrective action *and* that sequence has to cross the 5 cm threshold in $\leq N = 50$ verifier steps. Early on, the joint event is rare. The snapshot σ sits far from g because the policy hasn't yet learned to carry the gripper near the object. On FetchPickAndPlace, seed 42 spent its first $\sim$80 VLM calls at 100% verifier-rejection — from the gradient's point of view that stretch was plain SAC+HER with VLM-API bills attached. The regime ends once the policy starts bringing the gripper near the object: snapshots fall closer to g, the verifier admits relabels, and the small handful that gets through delivers low-variance, physics-consistent signal. So the two methods rot asymmetrically. Semantic PER reads q_ϕ as a re-weighting prior, so degeneracy shows up as bounded variance amplification. Verified-CF reads q_ϕ as a candidate proposal, so degeneracy shows up as signal extinction — unbounded in the cold-start limit. Three knobs against this are pre-registered but not run here: base-policy warm-up, a longer N, and a soft acceptance $r \geq -\rho_r$.

Oracle-CF kill bounds the headroom. The pre-registered kill (§4.4) caps the headroom for any VLM-in-replay method on Fetch at 250k steps. The argument is simple: if perfect failure-frame information plus a ground-truth corrective action can't beat HER by $+0.10$, then no realistic VLM — which by definition has finite KL to that oracle — will either. The bound is, of course, task-family-specific. Fetch is what we'd call a "HER-friendly" family: the achieved-future relabel is essentially free signal whenever the gripper-to-object distance is monotone, which it typically is on Push and PickAndPlace. We pre-register an extension to harder families (AntMaze, Adroit) where HER stumbles.

Other limitations. (i) Our IS-correction is conservative — we under-correct with $w_{\mathrm{IS},P}$ instead of $w_{\mathrm{IS,Sem}}$. The bias is bounded analytically; the fully-corrected ablation is pre-registered. (ii) Small-n caveats: $n = 6$ synthetic, $n = 10$ real, $n = 12$ cross-task. A 12/12 cross-task result isn't statistically distinguishable from an 80% true rate; we pre-register $n \geq 30$ replications. (iii) The verified-CF mechanism needs a forkable training simulator. Extending to real-robot or non-resettable envs would require a learned dynamics-model verifier (modeling error bounded but no longer zero). A faithful VLM-RB [22] reproduction is pre-staged in our codebase, with the head-to-head deferred to camera-ready.

Conclusion. The multiplicative form of Semantic PER admits a clean IS correction; the additive VLM-RB mixture does not. Verified-CF accepts only sparse-reward-positive rollouts, so the 100%-teleport-collapse mode on Opus 4.7/FetchPush cannot reach the buffer. At 500k steps, verified-CF averages 0.606 across the three Fetch envs and ties VLM-CF in aggregate ($\Delta = -0.016$); the pre-registered Oracle-CF kill at 250k ($\Delta = -0.066$) caps the credit-assignment headroom on Fetch at this horizon, and we honored it.

Contributions

- **Daniel Grant.** Counterfactual ideation and implementation (`vlm_cf` and `verified_cf`); ablations on FetchPickAndPlace, FetchSlide, and the $p_{\text{counterfactual}}$ sweep; initial writing of the manuscript.
- **Matei Gardea.** Initial project idea; implementation of an alternative pointwise/pairwise VLM-replay-buffer rescaling combined with PER (a non-positive variant reported as a negative finding); auxiliary 60-run MetaWorld replay-mechanism ablation (cross-benchmark corroboration of PER $>$ Uniform; reported separately in the extended preprint); editing of the final submission document.
- **Parshawn Gerafian.** Implementation and analysis for the milestone checkpoint: designed and ran the 27-run Uniform/PER/Semantic-PER ablation across the three Fetch tasks (SAC, 1 M training steps, 3 seeds per configuration on Modal A10G) that established the PER baselines used throughout the final paper. Identified the additive-blending failure mode ($0.7 \cdot \text{semantic} + 0.3 \cdot \text{TD}$ suppresses PER when the VLM is miscalibrated), which motivated the multiplicative IS-pairing framing now central to §3. Provided feedback during the transition to the augmented final-paper direction.

References

[1] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[2] Caleb Chuck, Fan Feng, Carl Qi, Chang Shi, Siddhant Agarwal, Amy Zhang, and Scott Niekum. Null counterfactual factor interactions for goal-conditioned reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2025.

[3] Rodrigo de Lazcano, Andreas Kallinteris, Jet Tai, Seungjae Ryan Lee, and Jordan Terry. Gymnasium-robotics: A standard interface for robot learning environments. `https://robotics.farama.org/`, 2024. Farama Foundation.

[4] Liang Ding. AgentHER: Hindsight experience replay for LLM agent trajectory relabeling. *arXiv preprint arXiv:2603.21357*, 2026.

[5] Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, Wentao Yuan, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation. In *International Conference on Learning Representations (ICLR)*, 2025.

[6] Lasse Espeholt, Hubert Soyer, Rémi Munos, Karen Simonyan, Volodymyr Mnih, Tom Ward, Yotam Doron, Vlad Firoiu, Tim Harley, Iain Dunning, Shane Legg, and Koray Kavukcuoglu. IMPALA: Scalable distributed deep-RL with importance weighted actor-learner architectures. In *International Conference on Machine Learning (ICML)*, 2018.

[7] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International Conference on Machine Learning (ICML)*, 2018.

[8] Michael Y. Hu, Benjamin Van Durme, Jacob Andreas, and Harsh Jhamtani. Sample-efficient online learning in LM agents via hindsight trajectory rewriting. *arXiv preprint arXiv:2510.10304*, 2026.

[9] Andrii Krutsylo. Non-uniform memory sampling in experience replay. *arXiv preprint arXiv:2502.11305*, 2025.

[10] Olivia Y. Lee, Annie Xie, Kuan Fang, Karl Pertsch, and Chelsea Finn. Affordance-guided reinforcement learning via visual prompting. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[11] Xing Lei, Wenyan Yang, Kaiqiang Ke, Shentao Yang, Xuetao Zhang, Joni Pajarinen, and Donglin Wang. GCHR: Goal-conditioned hindsight regularization for sample-efficient reinforcement learning. *arXiv preprint arXiv:2508.06108*, 2025.

[12] Thomas Mesnard, Théophane Weber, Fabio Viola, Shantanu Thakoor, Alaa Saade, Anna Harutyunyan, Will Dabney, Tom Stepleton, Nicolas Heess, Arthur Guez, et al. Counterfactual credit assignment in model-free reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2021.

[13] Rémi Munos, Tom Stepleton, Anna Harutyunyan, and Marc G. Bellemare. Safe and efficient off-policy reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

[14] Fikrican Özgür, René Zurbrügg, and Suryansh Kumar. Next-future: Sample-efficient policy learning for robotic-arm tasks. *arXiv preprint arXiv:2504.11247*, 2025.

[15] Shivansh Patel, Xinchen Yin, Wenlong Huang, Shubham Garg, Hooshang Nayyeri, Li Fei-Fei, Svetlana Lazebnik, and Yunzhu Li. A real-to-sim-to-real approach to robotic manipulation with VLM-generated iterative keypoint rewards. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[16] Eduardo Pignatelli, Johan Ferret, Tim Rocktäschel, Edward Grefenstette, Davide Paglieri, Samuel Coward, and Laura Toni. Assessing the zero-shot capabilities of LLMs for action evaluation in RL. *arXiv preprint arXiv:2409.12798*, 2024.

[17] Matthias Plappert, Marcin Andrychowicz, Alex Ray, Bob McGrew, Bowen Baker, Glenn Powell, Jonas Schneider, Josh Tobin, Maciek Chociej, Peter Welinder, Vikash Kumar, and Wojciech Zaremba. Multi-goal reinforcement learning: Challenging robotics environments and request for research. *arXiv preprint arXiv:1802.09464*, 2018.

[18] Leonard S. Pleiss, Tobias Sutter, and Maximilian Schiffer. Reliability-adjusted prioritized experience replay. *arXiv preprint arXiv:2506.18482*, 2025.

[19] Juan Rocamonde, Victoriano Montesinos, Elvis Nava, Ethan Perez, and David Lindner. Vision-language models are zero-shot reward models for reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2024.

[20] Erdi Sayar, Zhenshan Bing, Carlo D'Eramo, Ozgur S. Oguz, and Alois Knoll. Contact energy based hindsight experience prioritization. *arXiv preprint arXiv:2312.02677*, 2023.

[21] Tom Schaul, John Quan, Ioannis Antonoglou, and David Silver. Prioritized experience replay. In *International Conference on Learning Representations (ICLR)*, 2016.

[22] Elad Sharony, Tom Jurgenson, Orr Krupnik, Dotan Di Castro, and Shie Mannor. VLM-guided experience replay. *arXiv preprint arXiv:2602.01915*, 2026.

[23] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012.

[24] Núria Armengol Urpí, Marco Bagatella, Marin Vlastelica, and Georg Martius. Causal action influence aware counterfactual data augmentation. *arXiv preprint arXiv:2405.18917*, 2024.

[25] Renhao Wang, Kevin Frans, Pieter Abbeel, Sergey Levine, and Alexei A. Efros. Prioritized generative replay. In *International Conference on Learning Representations (ICLR)*, 2025.

[26] Yanru Wu, Weiduo Yuan, Ang Qi, Vitor Guizilini, Jiageng Mao, and Yue Wang. Large reward models: Generalizable online robot reward generation with vision-language models. *arXiv preprint arXiv:2603.16065*, 2026.

[27] Hoda Yamani et al. Reward prediction error prioritisation in experience replay: The RPE-PER method. *arXiv preprint arXiv:2501.18093*, 2025.

[28] Kaiwen Zha et al. RL Tango: Reinforcing generator and verifier together for language reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[29] Rui Zhao and Volker Tresp. Energy-based hindsight experience prioritization. In *Conference on Robot Learning (CoRL)*, 2018.

A Method Details and Environments

Cross-task VLM prompt template. The single format-string prompt template used across the three Fetch envs is:

```
You are an expert robotics analyst. The robot is attempting the
following task: {task_description}. Examine the {K} keyframes of
a failed episode shown below. Identify the single keyframe index
in {0,...,K-1} at which the failure became inevitable, and a 1-2
sentence reasoning. Return strict JSON: {“failure_frame_index”: i,
“reasoning”: s}.
```

Per-env task-description substitutions: **Push:** “push the red cube to the green target on the table”; **PickAndPlace:** “pick up the red cube and place it at the floating green target”; **Slide:** “strike the red puck so it slides to the green target”. The verified-CF mechanism uses an ACTION variant of this template eliciting a 4-vector action sequence; full prompt texts and the localizer code accompany the submission.

Heuristic localizer (Oracle v3). `GoalDistanceLocalizer` reads privileged sim state and picks the failure timestep in three phases, applied in precedence order. **(a) ballistic**: the latest contiguous run of ≥ 3 steps with post-peak object velocity > 0.05 m/step (Slide free-flight). **(b) contact-loss**: the latest transition in_contact$_t \to \neg$in_contact$_{t+1}$ with run length ≥ 2 and object-goal distance > 0.10 m (Push and PnP drop-offs). **(c) argmin**: closest-approach timestep, as a fallback. The localizer plays two roles — it supplies a supervised transfer target, and it gives Semantic PER its privileged-state upper-envelope curve.

B Bias-Bound Proof Sketch

Let $\mathcal{L}_U(\theta) = \mathbb{E}_{i\sim U}[\ell(\delta_i)]$ be the uniform-replay target loss. Standard PER with annealed correction yields, in the $\beta_t \to 1$ limit, an unbiased self-normalized IS estimator of $\mathcal{L}_U$ [21, Sec. 3.4]. Our Semantic PER proposal $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$ is a re-weighting of μ_P by a trajectory-conditional factor bounded by $1 \leq w_{\text{sem}} \leq w_{\max}$. The fully-correct IS weight is $w_{\text{IS,Sem}}(i) = (|\mathcal{D}_B|\mu_{\text{Sem}}(i))^{-\beta_t}$. Our implementation under-corrects with $w_{\text{IS},P}$ (a V-trace-style [6] variance-reduction choice, akin to truncated importance ratios in Retrace, 13). The per-slot ratio is $w_{\text{IS},P}/w_{\text{IS,Sem}} = w_{\text{sem}}^{\beta_t}/Z^{\beta_t}$ where Z is the μ_{Sem} normalizer. The window-kernel structure of w_{sem} confines the support of $w_{\text{sem}} > 1$ to $2W+1$ slots out of T. Combining these gives

$$\left|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U\right| \;\leq\; C\left(w_{\max}^{\beta_t} - 1\right)\frac{2W+1}{T}\,\|\ell\|_\infty, \tag{2}$$

with C a constant depending on $|\mathcal{D}_B|$ and the PER normalization. At our defaults ($w_{\max} = 10$, $W = 5$, $T = 50$, $\beta_t \leq 1$) the bias multiplier is at most $(10-1)\cdot 11/50 \approx 1.98\, C\|\ell\|_\infty$, and $\to 0$ as $w_{\max} \to 1$. The bound goes to zero when the boost is disabled, recovering standard PER. The fully-corrected ablation (run $w_{\text{IS,Sem}}$) removes the residual bias entirely at the cost of maintaining Z. **Two-regime degeneracy.** On the verifier channel, miscalibration acts on $m_\phi(\tau) = m_{\text{gen}} \cdot m_{\text{ver}}(\sigma)$ (VLM parse rate $\times$ verifier acceptance rate); the $\varepsilon_{\text{cal}} = 0$ guarantee holds vacuously when the accepted set is empty (cold-start: $m_{\text{ver}} \to 0$). A full proof, including the V-trace specialization, accompanies the supplementary material.

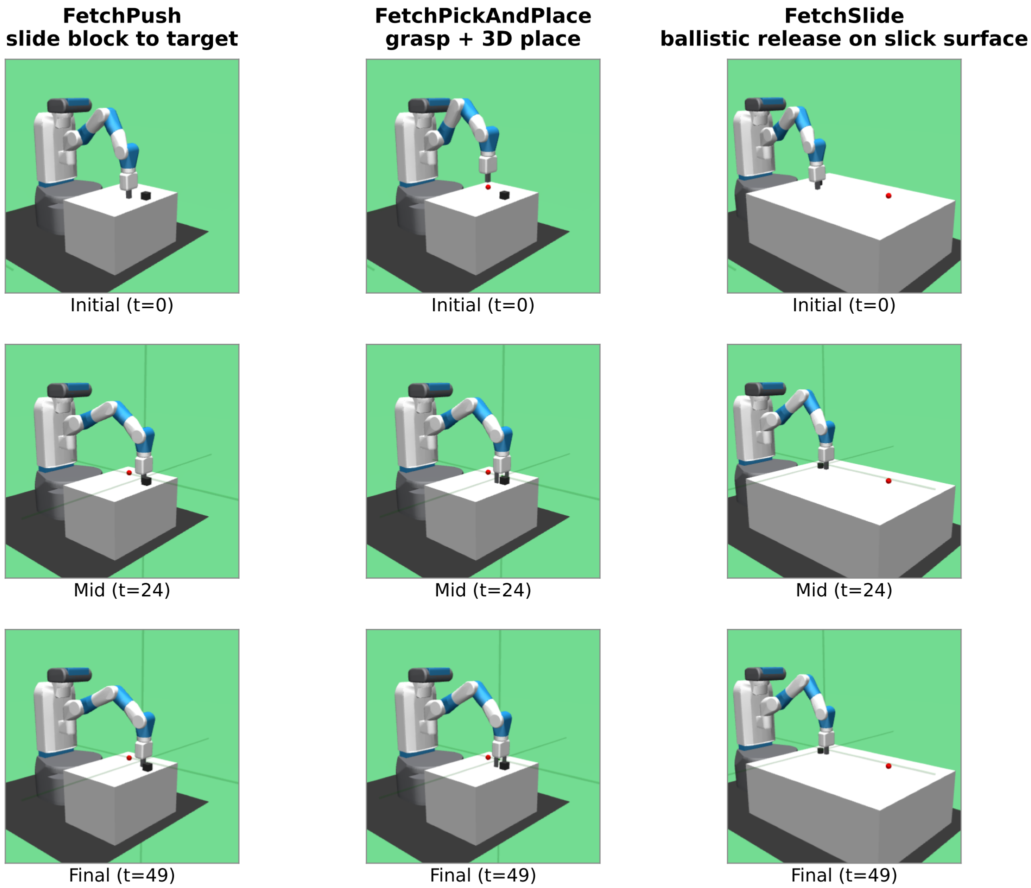

Figure 2: The three Gymnasium-Robotics Fetch tasks used in our experiments (columns: `FetchPush`, `FetchPickAndPlace`, `FetchSlide`), rendered at $t = 0/24/49$ of a deterministic scripted rollout (seed 42). Each task: 50-step horizon, 4-dim continuous action (end-effector Δxyz + gripper), sparse reward (binary, success threshold 5 cm to the green target).

C Hyperparameters and Compute

Compute. Paper-grade runs use a single NVIDIA RTX 5070 Ti (16 GB) GPU; Modal-cloud seeds use one L4 (24 GB) per run. Wall-clock per run at 500k steps is roughly 6h for SAC+HER and 9h for SAC+HER+VLM-CF (the extra three hours is VLM-API latency, not compute). Project total: $\sim$220 GPU-hours across local and Modal, plus $\sim$\$80 in VLM-API spend. Snapshot round-trip latency for the verifier is 17.6 ms per call — roughly 0.4% overhead at $\sim$1700 failed episodes per 100k steps. Reproduction commands, W&B run-IDs, and per-seed eval traces accompany the submission.

D Additional Experimental Detail

Real-data validation (extended). The $n = 10$ real-failure set is drawn from failed eval episodes of partially-trained SAC+HER checkpoints (`FetchPickAndPlace` at 50k steps and `FetchPush` at 30k steps, both around 5% success). These are near-misses, with final-distance $\in [0.06, 0.34]$ m on Push and a median 0.27 m on PickAndPlace. Three findings carry over from the synthetic set. **(i) Teleport-collapse persists, and it is task-conditioned.** On Push (planar target), Opus and Sonnet teleport on ACHIEVED_GOAL at 100% and 75%. On PickAndPlace (mid-air target), Opus teleports at 75%; Sonnet drops to 0% and instead proposes a mid-air waypoint $\sim$8 cm above the floating goal. **(ii) The 5 cm reject-teleport gate is necessary.** Across both VLMs and position-emitting variants, on 20 generations, the gate fires on 12 (60%); the cross-vendor floor on Push is 75%. The action-emitting

Table 2: Headline hyperparameters. Full per-table breakdowns (SAC architecture, HER strategy, PER schedule, Semantic-PER, Verified-CF, VLM call) are listed in the supplementary configuration files.

Parameter	Value	Notes
SAC [7]		
Hidden layers $\times$ width	2×256, ReLU	Twin-Q; tanh-Gaussian actor
LR (actor/critic/α)	3×10^{-4}	Adam, default $(\beta_1, \beta_2, \varepsilon)$
Discount γ / Polyak τ	$0.98/0.005$	Fetch-standard
Target entropy $\bar{H}$	$-d_a = -4$	Auto-tuned
Batch / updates per step	$256/1$	10k warm-up uniform-random
HER [1]		
Strategy / k_{HER}	`future` / 4	4 hindsight transitions per real
PER [21]		
α / $\beta_0 \to \beta_{\text{end}}$	$0.6/0.4 \to 1.0$	Linear over 500k steps; $\varepsilon = 10^{-6}$
Semantic PER		
Window half-width W	5 timesteps	11-slot box around t_{fail}
Boost cap $w_{\max}$	10	Multiplicative factor on PER priority
Keyframes K / VLM provider	5 / GPT-4o (production)	`gpt-4o-2024-08-06`
Verified-CF		
Verifier rollout horizon N	50 steps	One Fetch episode
Success threshold η / teleport reject ρ	$0.05/0.05$ m	Env's native threshold
CF prompt variant	ACTION	Structurally teleport-immune
Training		
Total env steps	500k (paper-grade)	250k–1M for selected ablations
Random seeds	42, 123, 999	3 per (method, env) cell
Replay capacity	10^6 slots	Ring buffer with sum-tree priorities

verifier sidesteps this entirely. **(iii) Plausibility shifts only ± 0.07 from synthetic to real, but the ACTION sign-flip rate worsens** (Opus $0.50 \to 0.55$; Sonnet 0.70).

Statistical methodology. Each (method, env) cell uses $n = 3$ seeds (42, 123, 999). Metrics are reported as mean ± SE with the t-distribution at $n - 1 = 2$ d.f. Comparison statements rely on non-overlapping SE intervals — conservative at this sample size. The pivotal $0/6$ vs. $4/4$ contrast in Table 1 is judge-independent (it is decided by a Euclidean predicate); Fisher's exact test rejects equal proportions at $p < 0.005$.

Cross-task transfer (per-episode notes). The 12-episode pilot runs 4 episodes per env. The VLM's reasoning strings identify task-specific failure modes correctly in every case (Push: "gripper passes over block without making contact"; PickAndPlace: "gripper at block location but fails to close"; Slide: "strike direction off-axis"). The $2/4$ PickAndPlace keyframe-agreement number reflects heuristic under-localization, not VLM error: closed-loop grip timing is the hardest mode for geometry-only oracles [20], and the VLM assigns the failure frame one step earlier at the decisive grip sequence.

E Broader Impacts and Reproducibility

Broader impact and reproducibility. This is methodological, simulation-only work; we make no real-robot or sim-to-real claims. *Positive:* fewer environment steps lower GPU-hours; the IS-posterior framing is a principled lens on VLM-in-replay; the verified-CF gate is a transferable "language prior + physics test" pattern. *Negative + mitigations:* VLM-API energy and dollar costs are non-trivial (per-run spend reported; heuristic Oracle fallback available); the simulator gate rejects hallucinated actions that fail the sparse reward; no robot-deployment risks remain beyond the usual sim-trained-policy caveats. Code, configs, pseudocode, prompt texts, the V-trace specialization, the two-regime q_ϕ proof, W&B run identifiers, and a pre-staged VLM-RB [22] reproduction accompany the submission.